Piramu-Mahalingam / DS-Course-Assignment-1

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DS-Course-Assignment-1 / Assignment 1.ipynb

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In [ ]:
```
#1. Difference between list and tuple
```

List: the apple is changed to orange

1. [] it is a ordered collection of multiple items stored in a single variable within square bracket **Example**: list1=['apple',1,34+9j,TRUE] 2.lists are mutable (Cheangeable) **Example**:list1[0]='orange'

3.It is used for collection that need to be updated such as append, delete and changing the value **Example**: Address Changes 4.it require additional memory for dynamic resizing and slower because of mutable

Tuple: throw an error

1. () it is a ordered collection of multiple items stored in a single variable within parenthesis **Example**: tuple1=('apple',1,34+9j,TRUE) 2.tuples are immutable (not Cheangeable) **Example**:tuple1[0]='orange'

3.It is used for fixed collections of items such as coordinates, date of joining, date of birth 4.performance is faster due to immutability and its require fixed memory.

In [ ]:
```
#2. compare the corresponding elements in two list
list1=[1,2,3,4,5,6]
list2=[6,5,4,3,2,1]
l3=len(list1)
for i in range(l3):
  if(list1[i]>list2[i]):
    print('list1 is greater than list2')
  else:
    print('list1 is less than list2')
```

```
list1 is less than list2
list1 is less than list2
list1 is less than list2
list1 is greater than list2
list1 is greater than list2
list1 is greater than list2
```

In [ ]:
```
#3.Write a Python user defined function that takes a list of numbers from user and returns a dictionary
```

```
 #with the number as the key and its square as the value

def squared_num(list1):
  for num in list1:
    sqdict={num:num**2}
    print(sqdict)
result=squared_num([1,2,3])
```

```
{1: 1}
{2: 4}
{3: 9}
```

In [ ]:

```
#4.Calculator program using user defined function
def calc(a,b,task):
  task1=task.lower()
  if(task1=='add'):
    print(a+b)
  elif(task1=='sub'):
    print(a-b)
  elif(task1=='mul'):
    print(a*b)
  else:
    print(a/b)


a=int(input('enter the integer value 1 '))
b=int(input('enter the integer value 2 '))
task=input('enter any one word with any case from this:add,sub,mul,div ')
calc(a,b,task)
```

```
enter the integer value 1 2
enter the integer value 2 5
enter any one word with any case from this:add,sub,mul,div ADD
7
```

In [ ]:

```
#5.Function for leap year
def leap_year(y):
  if(y%4==0 and (y%100!=0 or y%400==0)):
    print("Given year is leap year ")
  else:
    print("Given year is not leap year")
```

```
year=int(input('enter the year 4 digit '))
leap_year(year)
print('all leap year in 21st century')
start_year=int(input('enter the 21st century year -start '))
end_year=int(input('enter the 21st century year-end '))
for i in range(start_year,end_year+1):
  if(i%4==0 and (i%100!=0 or i%400==0)):
    print(i)
```

```
enter the year 4 digit 2024
Given year is leap year
all leap year in 21st century
enter the 21st century year -start 2010
enter the 21st century year-end 2024
2012
2016
2020
2024
```

In [ ]:

```
#6.pattern program
n=5
for i in range(1,n+1):
  print("\n")
  for j in range(1,i+1):
    print(j,end=" ")
```

```

1

1 2

1 2 3

1 2 3 4

1 2 3 4 5
```

In [ ]:

```
#7.conversion fron tuple to dictionary
tuple1=[("name","virat"),("surname","Kohli"),("yob",1986)]
dictionary=dict(tuple1)
```

```
print(dictionary)
```

{'name': 'virat', 'surname': 'Kohli', 'yob': 1986}

In [ ]:

```
#8.function should return odd and even indexed numbers
def fn_list(list1):
  l1=[]
  l2=[]
  l=len(list1)
  for i in range(0,l,2):
    l1.append(list1[i])
  print("odd indexed elements",l1)
  for j in range(1,l+1,2):
    l2.append(list1[j])
  print("even indexed elements",l2)




list1=[1,2,3,4,5,6]
fn_list(list1)
list2=input("enter the list elements , separeted ").split(",")
list2= [item.strip() for item in list2]  # Strip extra spaces
list2
fn_list(list2)
```

```
odd indexed elements [1, 3, 5]
even indexed elements [2, 4, 6]
enter the list elements , separeted 5,6,"hai",7
odd indexed elements ['5', '"hai"']
even indexed elements ['6', '7']
```

In [64]:

```
#9.find the Longest word
def longest_word(str):
  l1=len(str)
  for i in range(0,l1+1):
    list=str.split(" ")
    longword=list[0]
  for j in list:
    if(len(j)>len(list[0])):
```

```
      print("longest word is ", j)

str="python programming is fun"
longest_word(str)
str2=input("Enter the sentence")
longest_word(str2)
```

```
longest word is  programming
Enter the sentence"hai welcome"
longest word is  welcome"
```

In [66]:

```
#10 count the frequency of each element in a dictionary
list=[1,2,2,3,4,4,4,5,5]
counts={}
for num in list:
  if num in counts:
    counts[num]+=1
  else:
    counts[num]=1
print(counts)
```

```
{1: 1, 2: 2, 3: 1, 4: 3, 5: 2}
```